Avalon AI

A Decentralized Security Layer for Blockchain Transactions

Status: Early-stage research and prototype development
Seeking: Grants, Smart-contract security engineers, EVM developers, security researchers, and founding technical contributors

The problem

A blockchain transaction can be cryptographically valid and still be malicious. A compromised signer, deceptive transaction interface, dangerous contract approval, or misbehaving AI agent can initiate an action that follows the blockchain's rules but violates the owner's intent.

Most detection systems can identify suspicious activity, but detection alone does not guarantee that a transaction can be stopped before execution.

The proposed approach

Avalon AI is an open, community-governed transaction-security layer for compatible blockchain wallets and applications.

It combines deterministic security policies, transaction simulation, evidence-based threat analysis, and optional AI-assisted risk detection. The protected wallet—not an AI model or external analysis server—enforces the security policy.

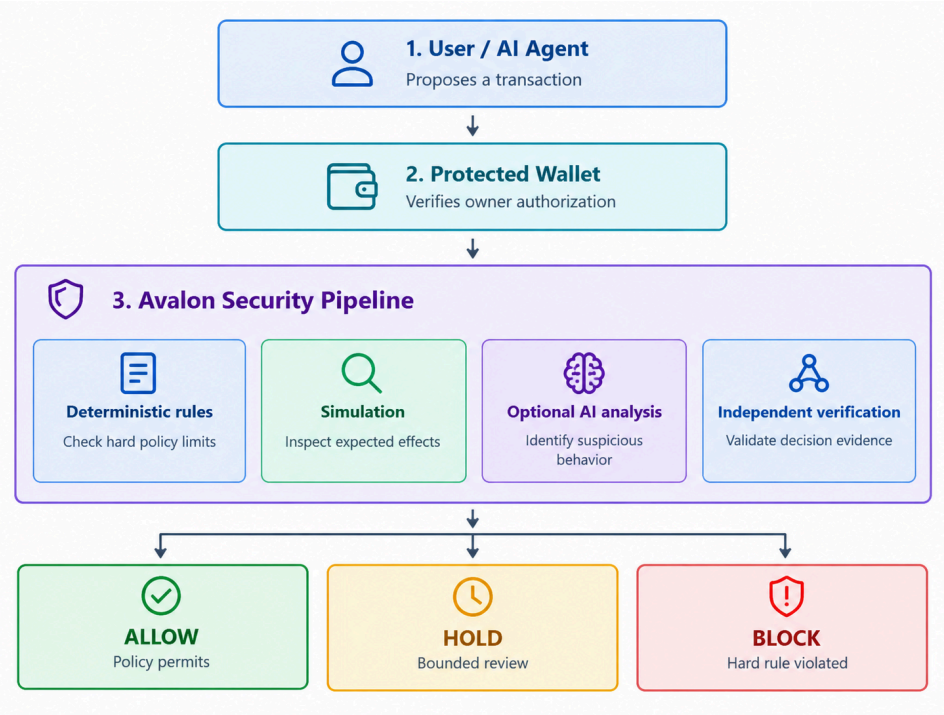
Core principle: AI may inform restrictions, but it cannot independently authorize transfers or override deterministic security rules.

Proposed architecture

1. **Authorize:** The protected wallet verifies the owner's signature and required permissions.
2. **Enforce policy:** Deterministic controls check spending limits, approved destinations, contract permissions, and sensitive administrative changes.
3. **Simulate:** A transaction simulator evaluates expected effects against a recorded blockchain-state snapshot.
4. **Analyze and verify:** Security rules and optional AI detectors identify risks. Independent verifiers reproduce checks where feasible.
5. **Enforce the outcome:** The wallet applies its configured policy:
    - **ALLOW** — required authorization and policy checks pass.
    - **HOLD / REVIEW** — a configured risk condition triggers a bounded, recoverable review path.

- **BLOCK** — an explicit hard rule or defined evidence-backed condition is violated.

Avalon cannot independently move funds. A compromised analysis service must not be sufficient to authorize a transfer or permanently freeze a wallet.



First technical milestone

Build a working proof of concept with:

- One EVM-compatible test network and one protected smart-contract wallet or vault.
- An initial deterministic policy engine and transaction simulator.
- A reproducible benchmark of malicious and legitimate historical transactions.
- A signed decision record and one independently operated verifier.
- Adversarial tests for forged, expired, replayed, and transaction-mismatched attestations, plus service outages and recovery.

No real-fund deployment is proposed until the enforcement design has undergone independent security review.

How we will evaluate it

Compare four protection levels:

- **L0:** Spending limits, permissions, and timelocks.
- **L1:** L0 plus deterministic security rules.

- **L2:** L1 plus transaction simulation and explicit invariants.
- **L3:** L2 plus AI-assisted risk analysis.

Measure detection and missed-attack rates, false positives, latency, cost, reproducibility, and the incremental benefit of AI. Publish methods, limitations, and results, including failures.

Seeking contributors

We need contributors with experience in:

- Solidity, EVM security, and smart-contract testing.
- Wallet architecture, account abstraction, and transaction execution.
- Backend security, cryptographic attestations, and independent verification.
- Adversarial testing, transaction simulation, and AI evaluation.

Project status and contribution

Avalon is an early-stage proposal, not an audited or production-ready security product. The architecture and its security claims must be validated through implementation, reproducible testing, and independent review.

We welcome critical feedback, threat-model reviews, prototype contributions, and reproducible adversarial test cases.

GitHub: https://github.com/thelennyjohnson/avalon-ai
Contribute: https://github.com/thelennyjohnson/avalon-ai/issues
Technical discussion: https://github.com/thelennyjohnson/avalon-ai/discussions
Contact: Telegram @mindvoyager

Avalon aims to make transaction-security decisions more transparent, enforceable, and independently verifiable—not to promise perfect security.